

DocStock

Reminder a day keeps Doc at bay

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Team Pearl



Isaiah Gamble

Isaiah Gamble is a senior at ODU majoring in Computer Science with a minor in Mathematics. He is a Coast Guard veteran and has "Experience Every Natural Disaster" on his bucket list. He plans on pursuing the arts after completion of his degree.

Julia Hairston is a senior at ODU majoring in computer science. She hopes to pursue a master's in data science in the future. She enjoys working out, listening to music, and doing puzzles in her free time.



Julia Hairston

Team Pearl



Shannon Stinnette

Shannon Stinnette is a senior at ODU majoring in Computer Science. While working part-time as a server, she is currently working towards a variety of networking certifications. Other than working she spends her free time enjoying video games, books, and hanging out with friends.

Anson Cheng is a senior at ODU majoring in Computer Science, hoping to pursue a Master's. He enjoys playing guitar, gaming with friends, and wracking his head on problems.



Anson Cheng

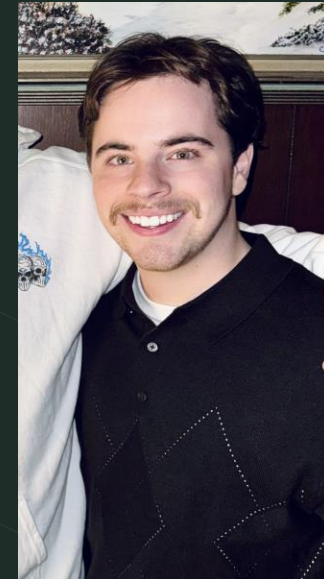
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Jessica Fischer

Jessica Fischer is a junior at Old Dominion University in the Linked Masters program for Computer Science. She enjoys gaming, working out, and hanging out with people in her free time.

Matthew Richards is a Senior Computer Science major at ODU. After graduating in May, he plans to move to Arlington, VA, and work for CACI doing backend development. He enjoys playing soccer, gaming, and playing the piano in his free time.



Matthew Richards

Team Pearl



Ab Aljazaeri

Abdelrahman "Ab" Aljazaeri is a senior at Old Dominion University majoring in Computer Science with a minor in Mechanical Engineering Technology. He likes to tinker with cars and dabbles in network security. He plans to keep working where software meets hardware after graduation.

Toni Sallaku is a senior athlete at ODU studying Computer Science. He is a swimmer, Lifeguard and Water Safety Instructor at the SRC. He likes working out, playing sports, and playing videogames.



Toni Sallaku



Problem Statement

Many people who take medications regularly have difficulty managing their medication schedules, remembering doses, tracking whether medications were taken, and knowing if their supply is running low. These difficulties can lead to missed, delayed, or repeating doses of medicine already taken.

Problem Characteristics

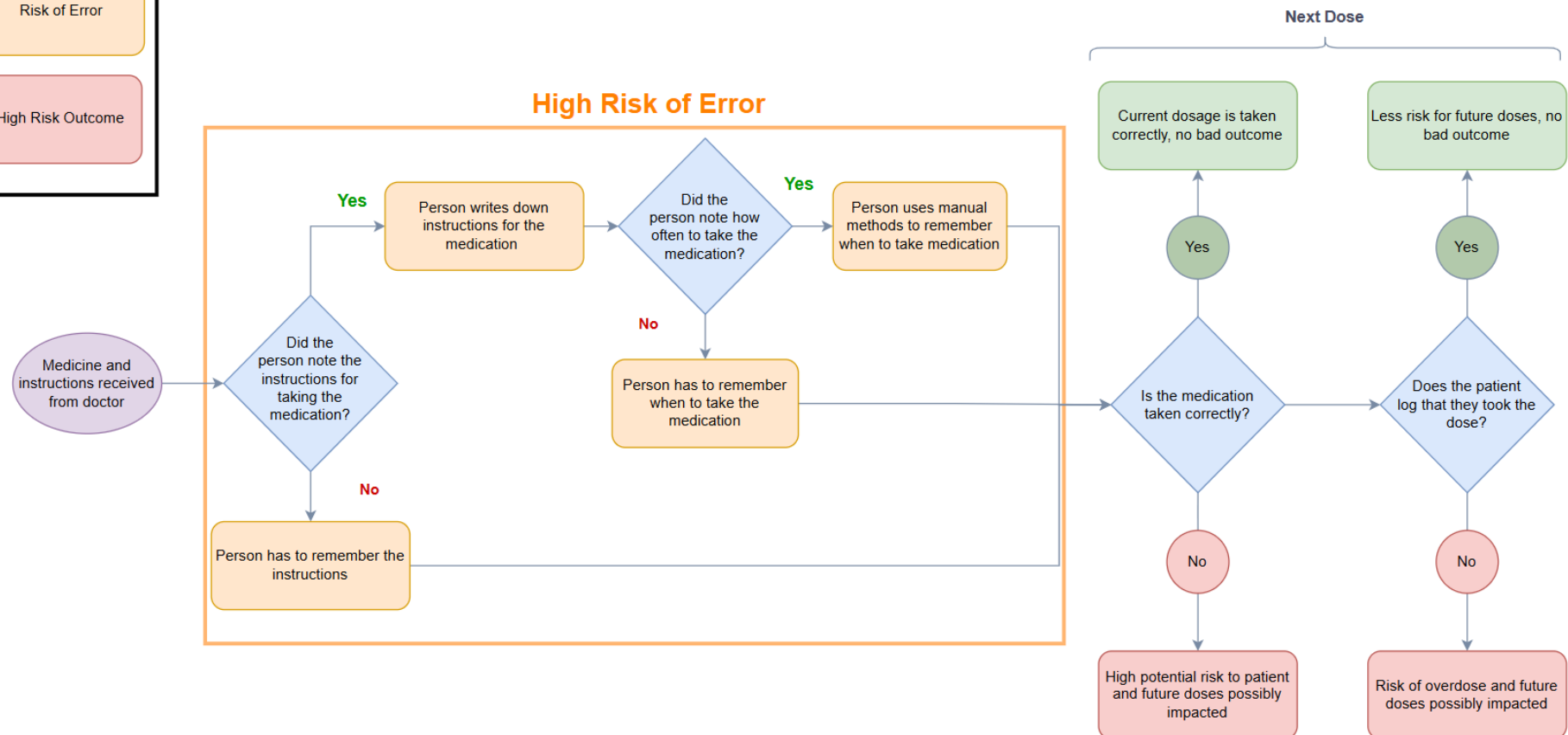
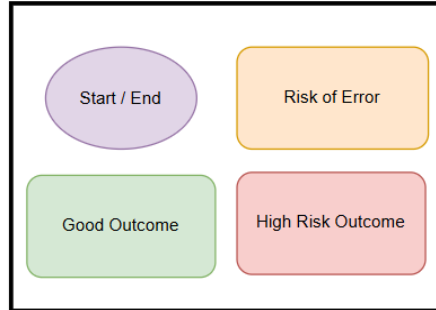
According to the Centers for Disease Control and Prevention (CDC), about one in five new prescriptions in the United States are never filled.

Approximately 50% of filled prescriptions are taken incorrectly with regards to timing, dosage, frequency, and duration.

Direct health care costs associated with nonadherence have grown to approximately \$100-\$300 billion of U.S. health care dollars spent annually.

Current Process Flow

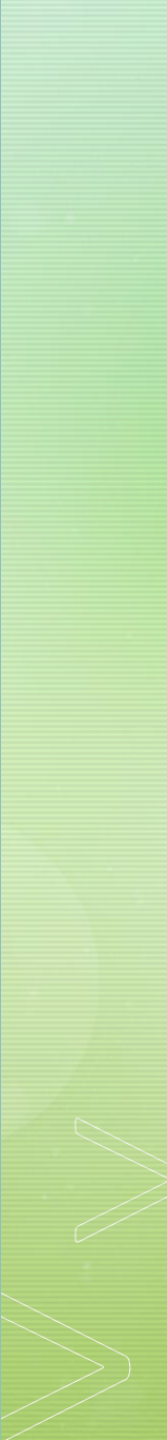
Key





Elevator Pitch

DocStock is an app designed to assist users with managing prescriptions, with our goal being increased adherence to medications.

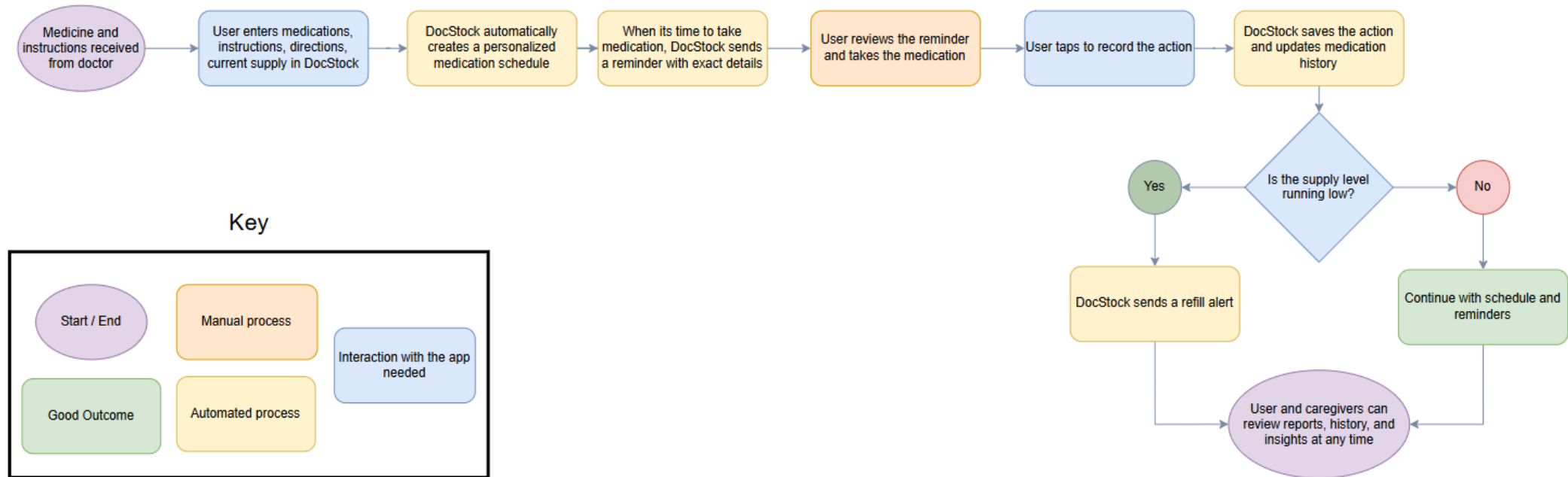




Solution Statement

DocStock will be a smartphone and web application that helps people manage their prescribed medication schedules. Users will input their medications and schedules, and the app will organize all this information into an easy-to-follow schedule. The app will provide reminders when medications are due to be taken, allow users to record their medication activity, and keep track of current supply of medicine.

Solution Process Flow



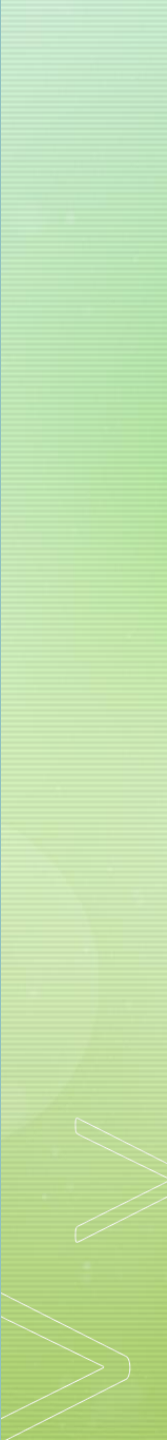
The number of manual processes is reduced to one compared to the current process flow.

What it Will Do

- Centralize every prescription in one dashboard reachable from any device.
- Schedule doses and send reminders so doses aren't missed, mistimed, or doubled
- Alert users before refills run out to prevent gaps in treatment
- Screen for drug interactions by checking each new medication against the user's active list
- Track adherence and export a provider-ready report to bring to appointments
- Share access with caregivers and support multiple profiles under one account



What it Will Not Do

- It will not automatically order refills for low medications.
 - It will not give medical advice or replace healthcare professionals.
 - It will not log symptoms, mood, or health metrics (blood pressure, glucose, etc.)
 - It will not integrate directly with pharmacies to pull prescription automatically.
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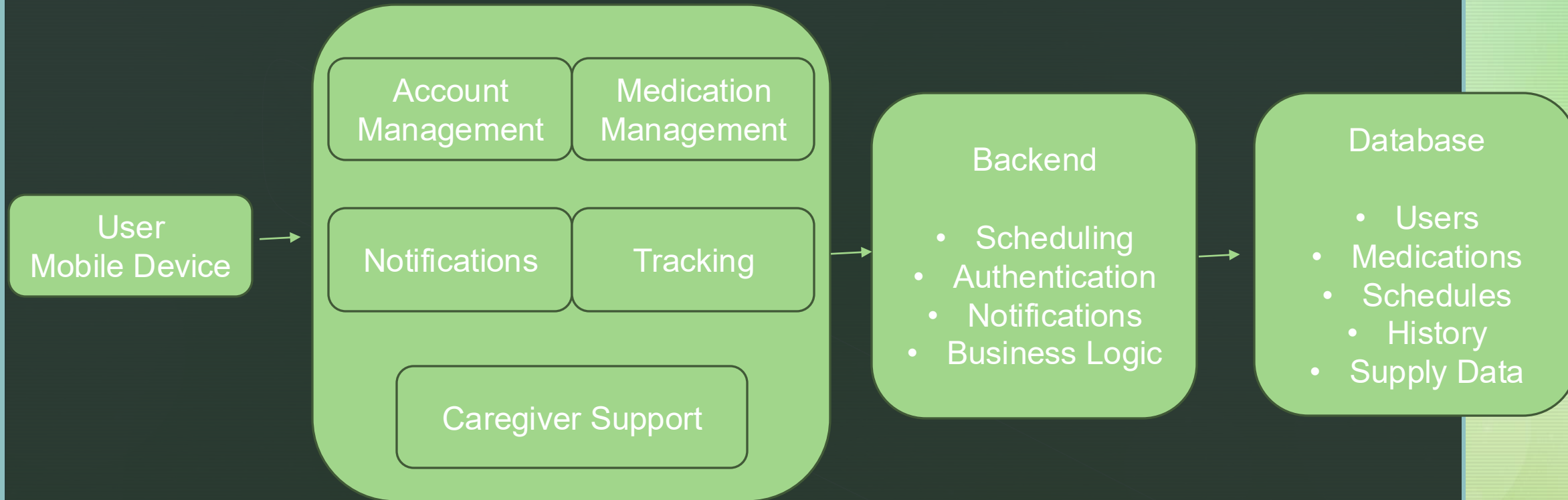
Competition Matrix

Feature	DocStock	Medisafe	MyTherapy	Round Health
Dose reminder & scheduling	✓	✓	✓	✓
Prescription refill alert	✓	✓	✓	✓
Drug interaction warning	✓	✓		
Caregiver / family sharing	✓	✓	✓	✓
Exportable adherence report	✓	Partial	✓	
Symptom / health metric logging			✓	
Multiple user profiles	✓	✓	✓	✓
Web-based (browser and phone app)	✓			
Free / no paywall	Partial	Partial	✓	Partial

Development Tools

- IDE: Visual Studio Code. Writing and editing code.
- Version Control: Git & GitHub. Managing code versions and team collaboration.
- Continuous Integration & Continuous Deployment: Automated testing and deployment.

Major Functional Components Diagram



Customer and End User Risks

Risk	Probability	Impact	Mitigation
User forgets to enter or update their medications	2	5	The app will use a simple set-up process with clear prompted instructions for adding and updating a medication. There will be a clearly visible button to add medication.
User ignores or misses medication reminders	3	4	Provide regular notifications and an easy way to record having taken a medication.
The app is difficult for some users to navigate	4	4	Large text, obvious buttons, simple navigation. User testing can identify problems before actual release.
User enters incorrect medication info	3	5	The app will require confirmation of medication details. It will clearly state that users should follow instructions provided to them by their healthcare providers.

Technical Risks

Risk	Probability	Impact	Mitigation
Delayed or failed notifications	2	5	Test notifications across supported devices. Provide in-app schedule for users to manually track. Alert users they must turn on app notifications to get them.
Smartphone compatibility problems	3	5	Test the application on multiple supported devices and OS versions.

Security Risks

Risk	Probability	Impact	Mitigation
An unauthorized user gains access to medication information	2	3	App will be password protected

Legal Risks

Risk	Probability	Impact	Mitigation
User claims the app caused harm because they relied on a reminder.	3	5	Communicate the app's limitations and advise users to verify medication instructions with their healthcare provider.
App unintentionally infringes on copyright or trademarks	2	3	App will use original or licensed materials with proper credit.

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Appendix

